



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2024/2025

THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE
DEGREE OF BACHELOR OF SCIENCE IN AGRONOMY,
BACHELOR OF SCIENCE IN ANIMAL SCIENCE, BACHELOR
OF SCIENCE IN ENVIRONMENTAL SCIENCE, BACHELOR
OF SCIENCE IN AGIBUSINES MANAGEMENT, BACHELOR
OF SCIENCE IN AGICULTURAL ECONOMICS, BACHELOR
OF SCIENCE IN HORTICULTURE AND BACHELOR OF
AGRICULTURAL EDUCATION & EXTENSION WITH
INFORMATION TECHNOLOGY

MAIN CAMPUS

**FIT 301: AGRICULTURAL AND ENVIRONMENTAL INFORMATION
SYSTEM DESIGN AND DEVELOPMENT**

Date: 4th February, 2025

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Do not write anywhere on this Question paper
- Answer Question ONE and any other THREE. Use examples where necessary

QUESTION ONE

- a. Briefly discuss the objectives of information system design and development as applied in environment or agriculture (5 marks)
- b. Explain the benefits of using system development life cycle in developing an information system (6 marks)
- c. Explain Gantt chart as a project management tool in reference to agricultural or environmental information design (2marks)
- d. Briefly discuss characteristic of an ideal system design appropriate for environment or agriculture (6marks)
- e. Using examples, explain different types of feasibility study as applicable to environmental or agricultural system design (6 marks)

QUESTION TWO

Assume you have been appointed as a system analyst for the researching, planning and recommendation of a software or system to be used in agriculture or environment, discuss the tasks you will perform to ensure the success of the system. (15 marks)

QUESTION THREE

Cost benefit analysis aids decision making for efficient allocation and use of resources. Discuss steps involved in the selection of the type of environmental or agricultural project to be undertaken (15 marks)

QUESTION FOUR

Policymakers need to have access to vital data and resources related to agriculture or environment to support decision-making processes. Discuss key platform features that can be used to facilitate collection and dissemination of the information. (15 marks)

QUESTION FIVE

Building an information system needs a model that provides a good stage for producing quality information system. Discuss types of system development models appropriate for agricultural or environmental system (15 marks)

QUESTION SIX

- a. Prototyping allows system analysts to get immediate feedback from the stakeholders. From a simple use of a prototype to web development, discuss the types of prototyping models (8 marks)
- b. Discuss areas covered during Post-Implementation Review in environmental or agricultural design (7 marks)